

Intelligent Ground Vehicle Competition 2026

AutoNav & Self-Drive



Cornell Electric Vehicles

Cornell University

Design Report

[Vehicle Name]

[Vehicle Photo / Sketch]

[Faculty Advisor's Signed Statement of Integrity]

Team Captain

Cam Mazzacane
clm357@cornell.edu · (585) 474-5070

Key Team Member

[Name]
[email@cornell.edu] · [Phone]

Faculty Advisor

[Name]
[email@cornell.edu] · [Phone]

[Author Name]

[Author Name]

[Author Name]

[Author Name]

[Author Name]

[Author Name]

[Author Name]

Contents

1	System Requirements and Design	1
2	Mechanical Design	1
3	Safety	1
4	Electrical/Electronic Design	1
5	Perception	2
6	Driving Logic	2
6.1	Extended Kalman Filter	3
6.2	Model Predictive Control	3
7	Key Performance Indicators	4
8	Analysis of Complete Vehicle	4
9	Cyber Security Analysis	5

1. System Requirements and Design

Things to Write

- Describe the system engineering process you followed to identify important System and Subsystem requirements.
- List at least 2 requirements for: (a) the mechanical design of your vehicle, (b) safety, (c) the electrical/electronic components
- For each challenge you are targeting (AutoNav and SelfDrive), list at least two requirements that guided your design for each of: (a) perception, (b) driving logic, and (c) key performance indicators.
- For each requirement you listed (12 for one course, 18 if both) explain what in these rules elsewhere that drives this requirement, describe how you will quantifiably measure the requirement, and set a target value.

2. Mechanical Design

Things to Write

- Describe the mechanical design of your robot, including CAD or other drawings of your vehicle
- Describe significant decisions on the frame structure, housing, and/or overall design
- Describe significant components of your vehicle's drive system, electrical power system, and the control system that enables computer-controlled motion. Components acquired ready-made must be identified, but their internal components need not be described in detail.
- Describe the significant mechanical components, including the frame, structure, suspension, housing, and weather proofing. Clearly identify the aspects of your mechanical design that are driven by each mechanical system requirement you listed.
- For each mechanical requirement you

listed, compare the target value with the most recently measured actual value, and discuss what this tells you about your vehicle

3. Safety

Things to Write

- Describe the robot's safety aspects while being transported, parked, and charging.
- Describe the robot's safety aspects while operating on the course(s), including the mechanical and wireless ESTOP systems.
- Clearly identify the aspects of your safety efforts that are driven by each safety requirement you listed.
- For each safety requirement you listed, compare the target value with the most recently measured actual value, and discuss what this tells you about your vehicle

4. Electrical/Electronic Design

Things to Write

- Describe significant power, electrical, and electronic components in your vehicle, including electronics, computers, sensors, and motor controllers
- Describe significant decisions made in designing/selecting the electrical and electronic components
- Describe the power distribution system's capacity, max. run time, recharge time, and safety
- Clearly identify the aspects of your electrical and electronic components that are driven by each electrical/electronic components system requirement you listed.
- For each electrical/electronic components system requirement you listed, compare the target value with the most recently measured actual value, and discuss what this tells you about your vehicle

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5. Perception**Things to Write**

- If you are targeting both challenges, you must include separate discussions of how your perception system matches each course.
- Describe processing of the sensor data to identify course features and items around the vehicle, including obstacles and lane markings, and the use of machine vision to perceive the environment.
- If targeting AutoNav, describe how the system identifies lanes and obstacles.
- If targeting SelfDrive, describe how the vehicle senses pedestrians, stop signs, and road obstacles.
- Describe the internal representation of the course built from the perception data which is used by the driving logic. This might be a sophisticated internal representation of the environment or a simple list of features. Include a discussion

of how it is created and updated as the robot moves.

- Clearly identify how your design of the sensor data processing system was driven by the perception requirements you listed.
- For each perception requirement you listed, compare the target value with the most recently measured actual value, and discuss what this tells you about your vehicle
- Developed a custom C++ Iterative Closest Point (ICP) node to process LiDAR point clouds for robust spatial mapping.
- Implemented a 3D-to-2D vertical slicing filter (using ROS 2 pointcloud iterators) to compress 3D LiDAR data, isolating course obstacles at specific Z-heights while filtering out ground and ceiling noise to prevent false obstacle detection.

6. Driving Logic**Things to Write**

- If you are targeting both challenges, you must include separate discussions of how your driving logic matches each course.
- Design of the lane following and obstacle avoidance systems must be specifically described, along with how the vehicle navigates through its environment. Describe how the system uses GPS for waypoint navigation and localization.
- If targeting AutoNav, describe how your robot uses the perception system's output and other information to follow lanes, detect and avoid obstacles, climb the ramp, and navigate to the GPS waypoints. Include how vehicle motion is generated and monitored to move the vehicle through the course, including how it deals with complex obstacles including switchbacks and center islands dead ends, traps, and potholes.

- If targeting SelfDrive, describe how the vehicle uses the perception system's output and other information to follow the rules of the road while navigating the course that includes pedestrians, stop signs, road obstacles, and other items.
- Clearly identify how your design was driven by the driving logic requirements you listed.
- For each driving logic requirement you listed, compare the target value with the most recently measured actual value, and discuss what this tells you about your vehicle

6.1. Extended Kalman Filter

Our vehicle makes use of a **custom Extended Kalman Filter (EKF)** to estimate the vehicle's state. For the general kinematics, the state is defined as:

$$\mathbf{x} = \begin{bmatrix} p_W \in \mathbb{R}^3 \\ R_{WB} \in SO(3) \\ v_W \in \mathbb{R}^3 \\ \omega_B \in \mathbb{R}^3 \\ \Omega_W \in \mathbb{R}^4 \end{bmatrix} \quad (1)$$

However, due to the group $SO(3)$ being a *non-Euclidean manifold*, this means that we cannot simply add or subtract orientation errors like the other states. This means that we need a local linear space, which is provided by the lie algebra $\mathfrak{so}(3)$, which is the tangent space of $SO(3)$. This allows us to define the error state as:

$$\delta \mathbf{x} = \begin{bmatrix} \delta p_W \in \mathbb{R}^3 \\ \delta \theta \in \mathfrak{so}(3) \\ \delta v_W \in \mathbb{R}^3 \\ \delta \omega_B \in \mathbb{R}^3 \\ \delta \Omega_W \in \mathbb{R}^4 \end{bmatrix} \quad (2)$$

This means that we can formulate the rotation update in error space as $R^+ = R_{\text{nom}} \exp(\delta \theta^\times)$ which makes use of the Rodrigues' formula (using $\exp(\delta \theta^\times) \approx I + \delta \theta^\times$ for small angle changes) to compute the

exponential map from $\mathfrak{so}(3)$ to $SO(3)$. This allows us to properly handle the non-linearities of the orientation state in our EKF, ensuring accurate state estimation for our vehicle.

Another unique aspect of the filter is that the F and G linearization matrices are computed using automatic differentiation via JAX. This is due to the complex dynamics formulated for the vehicle, which include non-linear slip and contact tire models, and full rigid body dynamics of the entire vehicle, including tire spin as exponential decay.

Combining all of these formulations, we can write out the predict step as:

$$\begin{aligned} \hat{\mathbf{x}}_{k+1|k} &= f(\hat{\mathbf{x}}_{k|k}, \mathbf{u}_k, \Delta t) \\ P_{k+1|k} &= F_k P_{k|k} F_k^\top + Q \end{aligned} \quad (3)$$

And the update state as:

$$\begin{aligned} \hat{\mathbf{x}}_{k+1|k+1} &= \hat{\mathbf{x}}_{k+1|k} + K_k(\mathbf{z}_k - h(\hat{\mathbf{x}}_{k+1|k})) \\ \beta &= I - K_k H_k \implies P_{k+1|k+1} = \beta P_{k+1|k} \beta^\top \\ &\quad + K_k R K_k^\top \end{aligned} \quad (4)$$

The full dynamics are formulated at [this repository](#). These are then also fed into a custom Model Predictive Control (MPC) model, which is detailed in the next subsection.

6.2. Model Predictive Control

Using the Jacobians formulated from the Extended Kalman Filter, we assemble the following prediction equation in error state:

$$\delta \mathbf{x}_{k+1} \approx F_k \delta \mathbf{x}_k + G_k \delta \mathbf{u}_k \quad (5)$$

where G_k is the Jacobian of the dynamics with respect to the control input. From this, we can stack the linearized dynamics over a horizon N into the state and control histories, defined as:

$$\mathbf{X} = \begin{bmatrix} \delta \mathbf{x}_1 \\ \vdots \\ \delta \mathbf{x}_N \end{bmatrix} \in \mathbb{R}^{16N}, \mathbf{U} = \begin{bmatrix} \delta \mathbf{u}_0 \\ \vdots \\ \delta \mathbf{u}_{N-1} \end{bmatrix} \in \mathbb{R}^{3N} \quad (6)$$

This is inserted into the standard lifted form, defined as:

$$\mathbf{X} = \Phi \delta \mathbf{x}_0 + \Theta \mathbf{U} \quad (7)$$

where:

$$\Phi = \begin{bmatrix} F_0 \\ F_1 F_0 \\ \vdots \\ F_{N-1} \cdots F_0 \end{bmatrix} \quad (8)$$

$$\Theta = \begin{bmatrix} G & & \\ FG & G & \\ F^2 G & FG & G \\ \vdots & & \ddots \end{bmatrix}$$

This formulation allows us to write the entire trajectory as a *linear function* of the initial error and the control input, which collapses the MPC problem into a single **quadratic program** (QP). We formulate the quadratic cost J as:

$$J = \mathbf{U}^\top (\Theta^\top \bar{Q} \Theta + \bar{R}) \mathbf{U} + 2 (\Theta^\top \bar{Q} \Phi \delta \mathbf{x}_0)^\top \mathbf{U} + \dots \quad (9)$$

Where $\bar{Q} = \text{blkdiag}(Q, \dots, Q, P_f)$ and $\bar{R} = \text{blkdiag}(R, \dots, R, P_f)$, and we omit the terminal cost weighting $\delta \mathbf{x}_N^\top P_f \delta \mathbf{x}_N$ for brevity.

Finally, this culminates in solving for the optimal $\mathbf{U}$ history via a QP as the following:

$$\begin{aligned} \mathbf{U}_{\text{opt}} = \underset{\mathbf{U}}{\text{argmin}} \quad & \mathbf{U}^\top H \mathbf{U} + 2 \mathbf{f}^\top \mathbf{U} \\ \text{s.t.} \quad & A \mathbf{u} \leq \mathbf{b} \end{aligned} \quad (10)$$

Where A and b are constraint matrices that allow us to control various aspects of the controller to fit our system. This is included in the same repository as above, and allows for simultaneous execution of both the localization and control frameworks into a single package. Furthermore, this allows us to have a robust control algorithm that requires little to no hand tuning of the controller, but rather tuning the dynamics themselves to ensure that the phys-

ical quantities are represented properly.

7. Key Performance Indicators

Things to Write

- If you are targeting both challenges, you must include separate discussions for each course.
- Describe how you will measure each of the key performance indicators you selected to check that your robot is competitive on the target course(s), what value/result you are targeting, and how you selected that value/result for the KPI.
- For each KPI requirement you listed, compare the target value with the most recently measured actual value, and discuss what this tells you about your vehicle

8. Analysis of Complete Vehicle

Things to Write

- Describe lessons learned during construction and system integration
- Identify key mechanical and electrical components that have failed during testing and discuss what changes you made to prevent repeat failures. Did your changes prevent repeat failures?
- Describe your software testing, bug tracking, and version control processes
- Describe your simulation-based testing process (e.g., Gazebo, etc.)
- Describe physical testing to date (mechanical, electronic, indoors, real world, etc.) and identify key differences in actual performance compared to predictions from simulation.

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9. Cyber Security Analysis

Things to Write

- Modern robotic and human driven vehicles are network-connected, contain large amounts of software from multiple vendors, and are an inviting target for cyber-attacks.
- List 3 cyber vulnerabilities present in your vehicle and discuss what steps should be taken to harden/remove them before it enters series production and is widely deployed.